Linkability of LLM Digital Twins: From Chance in 2023 to 60 % in an Open World, and the Control That Says When Not to Measure

Anonymous Author(s)

Abstract

Published LLM "digital twins" — per-respondent simulated answer vectors released alongside survey panels — are linkable to the humans they were built from, and that risk has a trajectory. Measured on three datasets and three generations of models, the twins of 2023 did not beat a demographic baseline recomputed on their own pool, and the twins of today exceed theirs. On the GPT-3 twins Argyle et al. released in 2023, the strongest designates the right respondent among 2,148 in **0.14 %** [0.05 ; 0.27] of cases against 0.10 % [0.02 ; 0.23] for that baseline — intervals that overlap, at a per-item accuracy *below* the modal answer. On twins built with recent models the same attack reaches **20.7 %** [19.0 ; 22.4] on Twin-2K-500 (2,058 respondents) in a closed pool and **4.28 %** [3.3 ; 5.6] in an open world at 1 % false accusations, against a human test-retest ceiling of 54.5 %. **Twin-2K-500 is the dataset that carries the claim**: armed identically on the same basin, a demographics-only comparator sits **29.9**× below the twin, and the gap survives giving a classical comparator the twin's own input. On the Park et al. archive the corresponding rates are **not evidence that the twin leaks beyond demographics**: that archive's "demographics-only" condition is conditioned on **eleven** attributes that leave almost every respondent unique in their own sample, and armed identically it comes within a ratio of **1.06**× of the agent (§ 5.8). **The three datasets differ in team, protocol, item count and pool, so "the risk grew with model capability" is confounded with "the protocols differ"; we state that confound in front rather than resolve it.** Inside a shared pipeline we report a twin-to-twin channel: two twins of the same person designate each other with no real human answer held by the attacker — 36.4 % [24.7 ; 56.5] top-1, clustered by configuration, at 60 common items. The "distance law" we preregistered — leakage falling with the number of differing elements — is refuted by our own measurements, and we claim no converse law; whether the channel survives between *independently built* pipelines is untested. Our control is not new — it is the prespend, gating form of Anonymeter's control baseline — and it costs nothing: no experimental arm is interpretable until its candidate pipeline beats a demographic baseline against the real humans. It stopped two of our own paid experiments and the Argyle replication. We also report what we could not establish. We preregistered that the observed coupling between imitation quality and leakage reflects an individual fingerprint; a null carrying only a matched per-person accuracy margin — which in fact leaks more than our own twin (31.6 % against 20.7 %) — reaches a rank correlation as high as ours (0.980, 5th–95th percentiles [0.951 ; 0.993], against 0.965 observed). Thirteen preregistered predictions were refuted, one is withdrawn, two are inconclusive and one untestable; our own twins failed to reproduce the leakage even with a paid frontier model; and the mechanism concentrating the effect on one item block remains unexplained. Finally, a 1998 mechanism (PRAM) applied to twins brings closed-world top-1 to **0.29 %** [0.10 ; 0.53] against the best attacker we measured — one who knows the mechanism and attacks the block it leaves untouched — at a measured cost of 4.4 points on inter-item correlations. It carries **no formal guarantee**: it republishes each item's within-segment histogram exactly, so its zero distribution and group errors are that republication, not a cost avoided.

Keywords

digital twins, large language models, re-identification, linkability, synthetic survey data, privacy

1 Introduction

Survey research is beginning to publish *digital twins*: for each human respondent in a panel, an LLM is conditioned on that person's profile and its simulated answers to the questionnaire are released as a per-person record. Toubia et al. published 2,058 such twins as Twin-2K-500 [41]. Park et al. built generative agents for 1,052 participants and cited privacy in restricting access to individual agent responses [30] — although the publicly posted replication package does carry those individual responses together with the agent conditions we analyse (§ 4.1, Ethical Considerations). The privacy question these releases raise has been named — by Park et al. themselves, by Bonagiri et al. [4], and by the AAPOR task force [35], which ranks synthetic response generation as the most risky of the core tasks it evaluates and names re-identification by linkage without quantifying it — but it has not been measured.

This paper measures it on three datasets spanning three generations of models, and reports both what the measurement supports and what it refutes.

1.1 The trajectory, and the confound that comes with it

Argyle et al. published in 2023 GPT-3 twins of ANES respondents, 12 matched items per person [3]. Attacked exactly as we attack the recent panels, those twins designate the right person among 2,148 in **0.14 %** [0.05 ; 0.27] of cases, against **0.10 %** [0.02 ; 0.23] for a demographic baseline recomputed on that same pool: the intervals overlap, at a per-item accuracy *below* the modal answer, and a nearest neighbour given the same eleven true answers carries **3.6 times more identity bits** (§ 5.8). Two years later, twins of the same kind of panel are re-identified at 20.7 % in a closed pool of 2,058 and 4.28 % in an open world at 1 % false accusations — and 29.9 times their identically armed demographic comparator in the closed pool (§ 5.8), where the 2023 twins do not demonstrably beat

Proceedings on Privacy Enhancing Technologies 2027(3), 1–18

https://doi.org/XXXXXXX.XXXXXXX

theirs. The risk was not measurable in the first published generation of twins; it is substantial in the current one.

The confound, stated before the claim rather than after it. The three datasets differ in team, protocol, item count (12 against 60 and 177), pool and questionnaire. "The risk grew with model capability" is therefore confounded with "the protocols differ". At matched **effective** information — 4.3 independent items on both sides, the same pool size, the same attack and the same tie-breaking convention — the twins of 2025 exceed their own demographic baseline (0.34 % [0.29 ; 0.40] against 0.13 % [0.11 ; 0.16]) while none of the three twins of 2023 exceeds theirs. Matching on the raw number of items is not enough: twelve Twin items still carry 46 % more independent information than the twelve ANES items, and the number of items inflates every attacker, the demographic baseline included, by a factor of 7.4 between 12 and 60 items (a second study of the same quantity puts it at 8.7). We therefore report a **direction, not a magnitude**: the 2023 endpoint fails our interpretability control and is not estimable (its ratio to baseline wanders between 1.4 and 3.7 across subsamples of its own pool), and relative to a nearest-neighbour attacker holding the same items the 2025 twin gains only 1.26 where the 2023 twin gains 0.96 — a difference we cannot separate from noise. Nor are the two ends in one metric: closed-world top-1 under a naive Hamming attack at the left, true detections at 1 % false accusations under A-LLR at the right, the second **undefined** at the left end, where its open-world AUC is indistinguishable from zero. These witnesses are post-hoc. Lifting the confound would need the same items, the same population and several generations of models with individually matched twins; no such resource is public (§ 7.4). We state the trajectory as a measured contrast between three published corpora, never as a causal effect of model capability.

One reading must be excluded. The Argyle result is **not a refutation of our scaling law**: it is out of domain. That law predicts the leakage of a twin that carries a person, and our interpretability control (§ 4.5) establishes that these twins carry none we can demonstrate. Taking 0.14 % for a refutation would commit, in reverse, the exact error the control exists to prevent — concluding from a measurement taken on noise (§ 5.8).

1.2 What we set out to show, and what happened

Our central hypothesis was that a twin's fidelity to *its particular person* and that twin's identifiability are expressions of one underlying axis. We preregistered a self-refutation test in two parts. The first passed: the rank correlation between imitation quality and leakage survives measuring the two axes on completely disjoint item sets (Spearman 0.969, 95 % CI [0.937 ; 0.993], 50 stratified random A/B splits of 60 items). The relation is not an artifact of recycled computation.

The second part failed, and against us. We constructed a *marginal null*: 100 artificial predictors matched to each person's own accuracy margin, with correct positions drawn at random without looking at the person — an object that is not structure-free, and that in fact leaks more than our own twin (31.6 % top-1 against 20.7 %). Adversarial review found a real defect in that first witness, since rebuilt without changing the verdict (§ 5.1). The rebuilt witness reaches mean Spearman **0.980**, 5th–95th percentiles [0.951 ; 0.993], against **0.965** observed over the 12 configurations; the observed value does not exceed it. Each null is compared only against the observed value measured on its own plan. **Reproducing the coupling requires nothing more than a matched per-person accuracy and correct positions drawn at random — prediction (b) is refuted.**

We report this as a result rather than a confession. A recent line of work argues that an attack on foundation models proves nothing until the null hypothesis is properly sampled: blind baselines routinely beat state-of-the-art membership inference attacks [9, 12], and a SaTML position paper makes the demand explicit [45]. The quality-leakage coupling reported in prose by Ward et al. [42] and Byun et al. [6], and contested by Platzer and Reutterer [32] and Adams et al. [1], has never been tested against such a control. We built it, and it absorbs our own effect. What survives is not a theory of why twins leak, but a set of measurements that do not depend on the coupling being individual-specific at all.

1.3 Contributions

(1) A trajectory measured across three generations of published twins. The same attack, under the same control, on twins released in 2023, 2024 and 2025: at their own demographic baseline then, 29.9× above it now in the closed pool (§ 5.8), with the protocol confound of § 1.1 forbidding any reading of it as an effect of model capability alone.

(2) A pre-spend stop rule, and what it cost us. The discipline is not ours: Anonymeter builds a control baseline into its risk estimate [19], and blind baselines demand the same of membership attacks [9, 45]. What is ours is the gating form, consuming no model call and applied before the first paid arm: **no experimental arm is interpretable until its candidate pipeline has been shown to beat the demographic baseline in top-1 against the real human answers** (§ 4.5). It stopped two of our own paid arms (§ 5.4), the Argyle replication (§ 5.8), and the `persona` condition of the Park archive — an agent built on a person's own self-description re-identifies her *less* well than demographics alone (0.26 % against 0.39 %).

(3) A twin-to-twin linkage channel within a shared pipeline, whose rate is not governed by how many pipeline elements differ. Two twins of the same person, sharing their persona source and differing only in model or output format, designate each other while the attacker holds no real human answer of any kind, on both Twin-2K-500 and the Park archive and against anti-artifact controls at or below 0.31 %; no number for this channel is reported without its common-item count. At one pipeline element changed, the rate runs from 17.2 % (model) to 81.4 % (decoding), and the *distance law* we preregistered — leakage falling with the number of differing components — is refuted by our own measurements (§ 5.4). Anonymeter's low linkability figure is not a counter-result, for the reason given in § 2.1. Whether the channel crosses *independently built* pipelines is untested: the arm meant to decide it is not interpretable (§ 5.4).

(4) The leakage is localised in the pattern of answers, not in memorised content. An ablation destroys the signal by permuting the order of a twin's answers while leaving their content untouched

(§ 5.5): what identifies is the dependence structure between answers. The pipeline also trains nothing on the target population, so no train/test gap exists to exploit, unlike Yeom et al. [44] and Feldman [15]; but our training-cutoff control lives in a regime where none of our twins identifies almost anyone (§ 5.7), so **we do not claim a measured dissociation between memorization and re-identification.** That narrower claim — a structural mechanism, localised by ablation — is what separates our channel from the memorization [8] and free-text inference literatures [26, 27, 38].

(5) The Narayanan–Shmatikov attack, instantiated on twins, with its human ceiling. Both moves are theirs [29]: the rarity-weighted likelihood score, and the out-of-sample criterion that drops the closed-world assumption and reports true detections against false accusations (§ 5.3, Figure 1). New here are the object, the pool, the human ceiling, and out-of-fold estimation of the weights. It corrects our own published Park figure upward — and raises that archive's demographic comparator further still, which is why Park no longer carries this paper's claim (§ 5.8).

(6) A defense measured with its real cost, and held against an adaptive attacker. Not a new mechanism — a variant of the 1998 Post Randomisation Method [21] — but applied to LLM twins with a risk-utility curve that is measured rather than assumed, tested against an attacker who knows how the defense works, and reported with what it does *not* protect: it offers **no formal guarantee**, and republishes the within-segment item histogram exactly (§ 6). We preregistered a comparison with differential privacy and **withdraw it**; we claim no superiority over differential privacy in either direction (§ 6.2).

(7) A bits-of-identity instrument that transports across datasets where the raw rate does not. A derived instrument, not a discovery: a case of Rényi min-entropy leakage [37], in the spirit of Eckersley's surprisal [14], addressing the transportability problem of the Rocher et al. scaling law [34] by another route (§ 5.6).

1.4 Negative results, stated in front

Thirteen preregistered predictions were refuted and one is withdrawn, two are inconclusive, and one could not be tested at all (Table 1, § 7.1); three later verdicts — the distance law (§ 5.4) and the third dataset's two predictions (§ 5.8) — sit outside that table for the reasons given there. Our own twins identify almost nobody, and the constructive demonstration meant to close that gap — raise fidelity with a frontier model and watch the leakage return — **failed**: `openai/gpt-4.1` on Twin's per-item recipe left fidelity at 0.1714 against their 0.708 (§ 5.7). We still do not know what the Twin-2K-500 recipe does differently. And the coupling that motivated this work does not survive its own control (§ 5.1). We consider a reader who stops here to have read the paper honestly.

2 Background and Related Work

2.1 Privacy of synthetic data

Stadler, Oprisanu and Troncoso show that generated tabular data remains linkable to its sources [39]. Giomi et al. formalise three attacks — singling out, linkability, inference — and conclude, on their datasets, that linkability is the weakest risk [19]. That conclusion is contested: Annamalai, Gadotti and Rocher [2] and Ganev and De Cristofaro [16–18] show reconstruction attacks defeating distance-to-closest-record metrics; Yao et al. [43] and Meeus et al. [28] generalise the critique; Golob, Pentyala and De Cock make these attacks an emerging standard [20]. Houssiau et al.'s TAPAS formalises attacker knowledge and the baselines an audit must beat [23].

What distinguishes us. This corpus concerns tabular microdata from classical generative models, evaluated by continuous similarity metrics. Our object is a vector of categorical answers produced by an LLM conditioned on a person, where no similarity metric substitutes for a direct matching attack.

2.2 Re-identification of real data

Narayanan and Shmatikov re-identify Netflix users [29]; de Montjoye et al. show four spatio-temporal points suffice on credit-card metadata [10]; Rocher, Hendrickx and de Montjoye model risk on incomplete samples [33] and extend it into a scaling law [34]. Taub et al. ground the statistical measurement of disclosure risk on synthetic data (CAP, TCAP) on the premise that synthetic generation breaks the link between identity and datum [40].

What distinguishes us. These works re-identify from genuine auxiliary data. Our attack's input is never a real datum of the target: it is a model output generated from a persona, compared against real answers the attacker separately holds. We falsify precisely the CAP/TCAP premise, on LLM-simulated survey microdata.

2.3 Privacy and LLMs

Carlini et al. document memorization and regurgitation of training sequences [8], and show memorization growing near log-linearly with model scale [7] — the leakage-versus-capability precedent closest to our own trajectory. Staab et al. infer personal attributes from innocuous free text [38]. Ko et al. de-anonymise online authors by multi-cue agentic inference [26] — the closest threatening neighbour — and Lermen et al. demonstrate the same attack at scale [27]. Theoretically, Yeom et al. connect membership attacks to overfitting [44]; Feldman shows memorizing rare cases is necessary for generalization [15]; tracing codes [5, 13] are the closest formal parent to our 1-in-N attack, bounding the worst case of an optimal adversary recovering an individual from aggregate statistics, though in dimension far exceeding the number of individuals.

What distinguishes us. Those works all start from free text bearing direct semantic cues, aggregated by an agent reasoning over it. Our channel is narrower and drier: closed-choice categorical answers, no free text, no agentic reasoning, and a mechanism localised in the dependence structure rather than in memorised content (§ 5.5). Unlike Yeom and Feldman, our pipeline trains no model on the target population.

2.4 Respondent simulation and twins

Argyle et al. launched the use of LLMs as panel substitutes [3]; their Dataverse archive supplies the 2023 end of our trajectory, and our measurement does not contradict their paper, whose study 3 compares matrices of Cramér's V — the association structure of a sample, not individual accuracy. Park et al. cite privacy in restricting access to their 1,052 agents' individual responses [30], while the publicly posted replication package nonetheless carries individual participant and agent responses (Ethical Considerations) — a risk anticipated, never measured. Toubia et al. publish the 2,058-twin panel we attack, treating neither privacy nor linkage [41]. Peng et al. document group-fidelity distortions on that same panel without connection to re-identification [31]. Bonagiri et al. issue a normative call to evaluate these risks without measuring them [4]. None of these works measures a re-identification rate from twin outputs. That is the gap we fill.

2.5 Closest prior art and contradictors

Ward et al. test 13 attacks over 9 generators and 48 tabular datasets, and state in prose a relation between quality and leakage close to ours [42]; Byun et al. observe a near-monotone frontier of the same type without a coefficient [6]. Three differences separate them from our claim: their quality is *aggregate* (MMD, Jensen-Shannon divergence, downstream AUC), never a correspondence to the right person; their risk is *membership* in a training set, not 1-in-N identification; and they publish no coefficient.

Three contradictors hold that fidelity and risk separate: Platzer and Reutterer [32]; Adams et al., for whom a generator without formal guarantees preserves fidelity and utility without evident privacy harm [1]; and Shafieinejad et al., observing decoupling as quality saturates within a single tabular diffusion model trained longer [36]. Our response is not that they are wrong: all three measure *aggregate* fidelity, never individual correspondence to the right person, and none tests whether the coupling they observe survives a null matched on per-person accuracy. Ours does not (§ 5.1). Their decoupling also unfolds over training time within one model, where ours is a cross-sectional correlation across twelve heterogeneous methods against 1-in-N re-identification.

2.6 Defenses

Our D4 is not new. Shuffling answers between units of the same segment is a variant of the Post Randomisation Method [21], which perturbs categorical variables through a known transition matrix leaving margins invariant in expectation, and of data swapping — the same family as the partially synthetic data tradition of Reiter and Drechsler, extended by Drechsler [11] and Bowen et al. [24]. What is new is the application to LLM twins with a measured risk-utility curve (§ 6).

3 Threat Models

We distinguish three scenarios, assuming the first one's strongest objection rather than deflecting it.

3.1 T1 — Panel holder publishes unkeyed twins; attacker holds the real answers

A panel holder releases per-person twin records without the key linking each twin to its respondent. An attacker who holds **the real answers to the same items** — an insider, a panel co-owner, a leak of the source file — matches twins to respondents. This is a *linkability* result in the sense of the Article 29 Working Party and the GDPR, not identification from public information: the quasi-identifier required is a specific questionnaire, rarely held outside the panel.

We state plainly what this is not. It does not show that a third party recovers a named person from public information. And on Twin-2K-500 itself the matching is trivial without any attack — `pid` equals `TWIN_ID`, and the twin files copy wave metadata line by line from the humans. The result therefore documents a generic risk for anyone releasing "anonymous" twins without that key, a practice Twin-2K-500 does not follow. The objection "this is not real re-identification, the attacker already holds the true answers" is assumed, not rebutted; our answers are the explicit model above and the open-world measurement of § 5.3. T2 below would remove the assumption entirely — but we did not manage to test it.

3.2 T2 — Two organisations publish twins of the same cohort; attacker holds nothing real

If several organisations each publish their own twins of the same panel with sufficient item overlap, a third party cross-references them holding no human answer at all. This threat model is named as an open problem by Jordon et al. [25] and covered by the A29WP linkability definition, but never measured: Anonymeter's linkability presumes real attribute values in hand [19], and Guépin et al. remove the real-auxiliary-data assumption but target membership in a single generator's training set, never matching between two independent generations [22].

We tried to test it directly, and could not: § 5.4's twin-to-twin measurement is same-team and same-persona-file, and the arm built as T2 actually describes fails the interpretability control of § 4.5. **T2 is neither supported nor refuted: it was not tested** (§ 5.4).

3.3 T3 — Pre-publication self-audit

A panel holder runs the attack against their own file before publication, to decide whether a defense is required — the constructive use of the method, and the one the artifact (Open Science) serves.

3.4 Attacker capabilities, all models

The attacker performs no training on the target population, has no query access to the generating model, and uses only rank-based matching over categorical vectors.

4 Data and Methods

4.1 Datasets

Twin-2K-500 [41], 2,058 respondents, public under CC BY 4.0. Target: wave 4, 60 items usable in common out of 108 (40 purchase items of the form "would you buy this product at this price?", 20 opinion items). Attacker context where applicable: 494 context items from waves 1–3 plus 14 demographic variables. Chance top-1 in the closed pool is 0.049 %. A human test-retest condition gives the empirical ceiling: 81.6 % [79.9 ; 83.2] top-1, accuracy 0.745.

Park et al. archive [30], 1,052 participants, 177 common items, replication package publicly posted. Of its five agent conditions we analyse four — composite, interview-only, survey-only, demographic — the `persona` condition being excluded by the interpretability control (§ 4.5). Human retest ceiling 96.8 %.

Argyle et al. archive [3], Harvard Dataverse, **CC0 1.0**: 4,270 ANES 2016 respondents, of whom **2,148** answer all **12** matched items; chance top-1 is 0.047 %. Each twin column is a GPT-3 (`davinci`) prediction produced while holding that item out and supplying the person's eleven other **true** answers, so these twins are *better* informed than our persona twins, not worse: that confound runs against us. Three twin versions of the same people exist (main decoding, temperature 0.01 and 1.0); they vary decoding, not epoch, and we never present them as survey waves.

No new data was collected. The attack uses only already-published content.

4.2 Methods compared

Twelve heterogeneous methods carry the quality-leakage analysis: 8 LLM twins and 4 statistical reference points (Demographics Only, B1 argmax over 14 demographics, B2 argmax, PMM k=10). Additional comparators appear in § 5.2: logistic regression on context, a k=1 donor on context, two individualised *generators* (sampled LR, Gaussian copula), and an explicitly advantaged k=1 donor fitted directly on the target block. Sequential CART and a 300-tree forest were tried before that donor and abandoned at 0.45–0.53 accuracy: the retained comparator was not chosen to lose. On the Argyle archive the comparators are **B-demo**, a leave-one-out imputer on the four demographic items, and **B-oracle**, the same imputer given the person's eleven other true answers — the comparator matched to that archive's own conditioning.

4.3 Metrics

Attacks. Two matching rules are used throughout, and every rate is labelled with the one that produced it. The **naive** attack ranks candidates by Hamming-type agreement over common items. The **strong** attack (A-LLR) instantiates the Narayanan–Shmatikov rarity-weighted log-likelihood [29], whose parameters we estimate **out-of-fold over 5 folds**. A third variant (A-MI) scores higher still on the Park archive (93.25 %) but estimates its weights **in sample**, an advantage declared in the preregistration, so we never report it as our result. An **adaptive** attacker, used only against the defense (§ 6.1), additionally knows the mechanism and which items it leaves untouched.

Closed-world top-1 and top-10: rank of the true target among all N candidates by the rule stated. *Open-world*: true-positive rate against false-positive rate, where the system may decline to accuse (§ 5.3). *Bits of identity*: log2 of the ratio of correct one-shot guessing probabilities before and after conditioning on the persona — a Rényi min-entropy leakage quantity [37]. *Imitation quality*: the share of the human test-retest floor attained.

Unless stated, intervals are 95 % bootstrap confidence intervals resampling persons, not methods: an interval on a quantity computed across the twelve methods is conditional on those twelve and does not generalise beyond them. Intervals that resample something else — sub-pool or tie-breaking draws, nulls — are not confidence intervals on the rate, and are labelled where they appear.

4.4 Preregistration and self-refutation

Attack, defense, mechanism, the two archive replications, the disjoint-item test and the distance law were each preregistered before computation, and we report every refuted prediction in Table 1. A census found **no inversion** between a plan and its result across the 28 verifiable pairs; for the large majority the ordering is provable from version-control history, while a small number were committed alongside their results or left unversioned; we state this as a limit rather than claim an unbroken chain. The plans are not yet third-party timestamped.

4.5 Validity controls

Provenance. On all three archives we verified that the attacker's context cannot contain the attacked answers and that no twin copies an earlier wave: symmetric wave-following on Twin-2K-500, whose context carries no wave-4 column; on the Park archive a wave-1 bias shared by the `demographic` condition, hence a calibration artifact and not item leakage; and on the Argyle archive item-by-item hold-out, for the twins as for B-demo and B-oracle. Counts and rates: the appendix.

Empty-run validation. Uniform ranks yield 0.00 bits; the B0 mode baseline yields −0.004 [−0.003 ; 0.002] bits.

Interpretability control, adopted after it cost us an experiment. We state as a rule the control whose absence invalidated two of our own arms (§ 5.4): **no experimental arm is interpretable until its candidate pipeline has been shown to beat the demographic baseline in top-1 against the real human answers**, on the pool and items actually attacked, with the baseline recomputed on those same rows rather than carried over as a constant. The check consumes no model call and must precede the first paid contrast; a pipeline that fails it yields twins whose contrasts oppose noise to noise and return chance whatever is manipulated. It has since excluded our own pipeline B, all three GPT-3 twins of the Argyle archive (§ 5.8), and the Park archive's `persona` condition (§ 1.3). Baselines are always taken at the pool size of the arm they judge, being strongly pool-dependent (§ 5.4); **and a baseline is only as weak as the attributes it is given** — the Park archive's is not, in that sense, a demographic baseline at all (§ 5.8).

5 Results

5.1 The coupling, and the null that absorbs it

Across 12 heterogeneous methods, the ordering by imitation quality and the ordering by re-identification rate nearly coincide. Three rank correlations appear for that statement, and they are three measurement plans, not three estimates of one number. **0.958** [0.930 ; 0.993], fidelity taken as raw accuracy on whole items. **0.969** [0.937 ; 0.993], the preregistered disjoint-item test over 50 stratified A/B splits of the 60 items, 100 % of them above 0.7 — prediction (a) confirmed, the relation is not an artifact of recycled computation. **0.965**, fidelity on Figure 2's axis (`fidelite_plancher`), the value the null below is compared against. Both intervals resample **people within each configuration** and hold the 12 configurations fixed; they are conditional on this set of methods and do not generalise. Resampling the **configurations** — the unit the claim is about — widens that interval substantially without changing its sign, and a permutation test rejects independence; these witnesses are post-hoc, and both intervals are reported in full in the appendix.

Prediction (b) was refuted, and the witness that refutes it has since been rebuilt to remove a real defect without changing the verdict: its *wrong* cells avoided the person's true answer (`c7_disjoint.py` l. 133), inflating its leakage by a factor 1.3 (41.2 % against 31.6 %

once wrong values are drawn from the population marginal). Corrected, it still reproduces the coupling — **rho 0.980, 5th–95th percentiles [0.951 ; 0.993], against 0.965 observed** over the 12 configurations, 100 replicates as preregistered (Figure 2) — so **prediction (b) remains refuted**. It is not an information-free object, and we do not describe it as one: at matched accuracy it leaks **more** than our twin (31.6 % top-1 against 20.7 %).

We therefore do not claim, anywhere in this paper, that a twin's fidelity to its person causes its leakage, nor that fidelity and identifiability are one axis, nor that at equal accuracy every twin leaks more than every predictor: the residuals of the regression on accuracy do **not** separate the families, 3 of 7 LLM configurations falling below the line. One positive point remains testable outside this circularity: the human retest dominates the best twin on both axes — 0.303 against 0.154 and 0.520 against 0.067 on the disjoint-split scale, 81.6 % against 20.7 % of closed-world top-1 on Figure 2's — so there is no saturation at the top of the range. The two scales are not interchangeable and nothing here mixes them.

Limits. Raw accuracy is only a noisy proxy (r = 0.72, p = 0.008) for the right axis, and Spearman is coarse at n = 12 — the null touches 1.000 through small-sample effects — so the verdict concerns the order of magnitude, not the third decimal.

5.2 Closed-world rates and the comparator question

At comparable marginal accuracy, no non-LLM predictor we tested comes near the twin: the strongest of them reaches **0.45 %** top-1 (the persona-conditioned comparator of the next paragraph) where the twin reaches 20.7 % [19.0 ; 22.4] under the naive attack and **23.23 % [21.5 ; 25.0]** under the strong one — **in a closed world, with the target always present in a pool of 2,058**. The comparator figures are naive-attack rates, which makes the contrast conservative. JSON Persona GPT4.1 attains that rate at accuracy 0.590; the six comparators, at accuracy 0.457–0.511, all sit at or below **0.25 %** top-1 — among them two individualised *generators*, sampled LR and Gaussian copula, which answers the objection that we oppose an individualised generator to predictors that regress to the mean. Each rate with its interval: the appendix.

At strictly equal input information the gap holds, and that is the form of the claim we defend. Given the 634 wave-1–3 columns the persona is made of — from which the 60 attacked items are absent — the best classical generator conditioned on the individual reaches **0.45 %** [0.20 ; 0.76] against **20.66 %** [19.0 ; 22.3] for the LLM twin, both terms from one run. A twin conditioned on the **demographic segment alone**, which has never seen the individual, still reaches **2.15 %** [1.57 ; 2.77] — **4.7 times** that best classical comparator, itself fed the individual. The gap is therefore not attributable to the input information. *(Post-hoc, not preregistered.)*

We then gave a comparator an explicit advantage: a k=1 nearest-neighbour donor fitted **directly on the target block's answers**, seeing what the LLM twin never sees. It reaches accuracy 0.584 against the twin's 0.590, carries genuine individual signal (0.584 true against 0.442 under within-segment permutation), and despite that obtains a top-1 of **0.00 % [0 ; 0.18]** (Clopper-Pearson, n = 2,058).

This claim does not generalise to every leakage metric, and top-10 runs the other way. That same advantaged comparator reaches a top-10 of **53.2 % [51.0 ; 55.3]**, *above* the twin's 42.7 %: the direction of the whole contrast depends on the choice of k, which no principle fixes at 1. We claim neither that all individualised prediction identifies nor strict accuracy equality — the generators plateau at 0.476. Neither synthpop/CART nor CTGAN was tested.

5.3 Open world — the defensible measurement

Removing the closed-world assumption gives the number we consider hardest to attack in review. At a false-accusation rate of 1 %, the strong attack (A-LLR, parameters estimated out-of-fold over 5 folds) recovers the right person **60.17 % [54.5 ; 64.4]** of the time on the Park archive and **4.28 % [3.3 ; 5.6]** on Twin-2K-500, under a person-level bootstrap that **re-estimates** its parameters in each resample, every copy of a person staying in one fold. At FPR = 0.1 % the quantity is **not identified by the data** — the threshold there rests on one absolute false positive on Park and two on Twin, and hundreds of admissible thresholds span almost the whole range of attainable detection rates — so **we report no point estimate at that FPR**. At FPR = 1 % the same construction is stable, and **the headline figures are the ones at 1 % FPR**; the threshold census establishing both statements is in the appendix. The human ceiling at FPR = 1 % is 54.5 % [50.8 ; 57.5] and 90.7 % [88.2 ; 93.3].

Statistical comparators stay indistinguishable from noise at both FPRs on Twin-2K-500 — and only there. Replayed under the same strong attack as the target, Twin's demographic comparator gives 0.00 % TPR at 0.1 % FPR and **0.0486 %**, one person, at 1 % FPR, below its own naive value, so the 4.28 % stands clear of it; PMM k=10 stays at 0.00 % on both archives under both attacks. On the Park archive the demographic comparator is **not** at noise once armed with the same attack — 29.09 % at 1 % FPR — which is why that archive no longer carries this paper's claim (§ 5.8). These witnesses are post-hoc.

These figures replace the ones we first published, and the correction runs in both directions. The naive attack gave 3.04 % and 20.39 % at 1 % FPR. On Park the strengthening is statistically clear at that FPR — no overlap, the rate **triples**. **On Twin-2K-500 the apparent gain does not survive its interval**: 3.04 → 4.28 % overlaps broadly, so we do not present the strong attacker as identifying better than the naive one here.

Our preregistered prediction was refuted, and one half of it then reversed by a better attack (> 5 % Twin, > 30 % Park; Table 1, row 2).

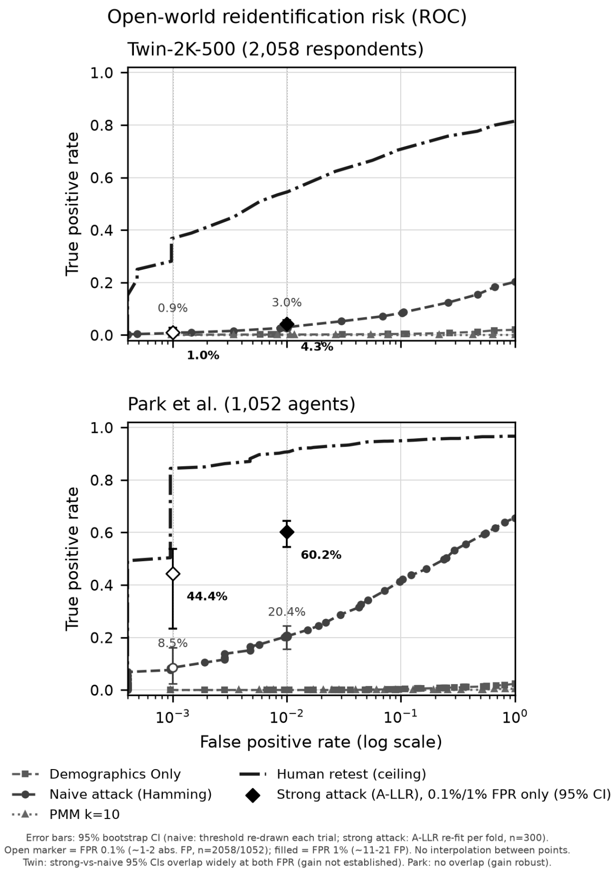

Figure 1: Open-world detection, naive and strong attack. Two panels, one per dataset, false-accusation rate on a log scale against true-detection rate. Each overlays the naive Hamming attack as a full ROC curve with the strong A-LLR attack as isolated diamonds at two thresholds only (FPR = 0.1 % and 1 %): no curve should be read into the diamonds. On Twin-2K-500 comparators sit near zero; on the Park archive the armed demographic comparator does not (§ 5.3). The human retest ceiling sits above. *At a glance*: at a tenable false-accusation rate the risk is real and, on Twin-2K-500, far above the comparators under both attacks, while on Park the strong attack lifts the agent and its own comparator together. Data: `resultats/c7-monde-ouvert-roc.csv`, `resultats/c7-attaquant-fort.csv`.

5.4 The twin-to-twin channel within one pipeline, what governs it, and why our T2 test does not count

Two twins of the same person designate each other when they share enough items, with no real data on the attacker's side. The measurements below share one team and, for Twin-2K-500, one persona source per person: they gauge the channel inside a pipeline, not T2 (§ 3).

Twin-2K-500, tabulated by common-item count. 30 pairs sharing 60 common items: mean top-1 **36.4 %**, 95 % CI **[24.7 ; 56.5]** clustered by configuration — the independent unit is the configuration, not the pair, as elsewhere in this paper: the 30 pairs come from 6 configurations. Individual pairs run from **11.9 %** to **83.6 %** (SD 18.6 points), so the mean should not be read as a typical value. 12 pairs sharing 19 common items: mean top-1 **0.45 %** (chance 0.05–0.10 %). The anti-artifact control, a decoy from the same segment, stays at or below **0.10 %** on both item regimes (the appendix).

Park archive replication: 1,052 agents, 177 common items, top-1 **11.9 % [10.1 ; 13.8]** and **13.0 %** in the two directions, against a decoy control **at most 0.31 %**. **Preregistered prediction refuted**: we set a threshold of at least 20 % top-1 there and 11.9–13.0 % does not reach it, though the preregistered *failure* criterion is not met either, at 5–8× the demographic bound (the appendix).

We never report a single figure for this channel without its common-item count; a previously circulated average of 26.11 % pooled the two item regimes and is withdrawn. Nor do we claim leakage crosses waves: that arm is infeasible, with 0 items in common.

The channel is not an artifact of a shared prompt template or of demographics. Decomposing agreement between two twins of one person over the 60 common items, the person contributes **82 %** of the rise above the floor between *strangers*, the demographic-ideological segment the rest; a pure-population witness identifies the right person **0.07 %** of the time against **36.4 %** for the real attack (the appendix).

The cross-dataset anomaly, explained in part. Park shares three times Twin's items and leaks roughly three times *less*, and neither item count nor pool size accounts for it: the 177 GSS items are strongly redundant, giving ≈ **9.0 effectively independent items against** ≈ **10.1** on Twin, and shrinking Twin's pool to Park's *raises* its top-1 rather than lowering it. **A residue of roughly 3× remains unexplained**, the remaining candidates — persona format and length, and the generating pipelines — not testable on what we hold. The same denominator places the third dataset far below both (§ 5.8; the appendix).

Limits. The 12 low pairs are exactly those at 19 common items, so the effect follows the shared-item count rather than an aberrant configuration. On Park the item-rate relation does not follow Twin's.

T2 itself: two genuinely independent pipelines, and a test that does not count. Two pipelines sharing nothing but the target person — `deepseek-v4-flash` JSON dossiers against `qwen-2.5-72b` narrative biographies — were matched on the same 60 items (n = 142 of 200 planned, stopped by a transport error), giving top-1 1.76 % [0.35 ; 3.63] against a demographic baseline of 9.2 %. **We no longer read this as a refutation.** Neither arm produces twins that identify the real person better than chance on that pool, where the Twin team's own twins reach 20.2–38.9 % (the appendix), so the comparison opposes two noise sources: it measures our instrument, not the world. The 36.4 % above, a same-pipeline measurement, is not evidence for T2 either, so T2 stands open both ways.

Which element differs, not how many — and a preregistered law our own measurements refute. Among the six Twin-2K-500 configurations that pass the interpretability control (2,058 people, 60 common items, symmetric attack), changing a single published pipeline descriptor gives **81.4 %** for decoding alone, **67.1 %** for the prompt template, **33.5 %** for reasoning and **17.2 %** for the model. We preregistered a *distance law* — leakage falling with the number of published descriptors that differ — and our own measurements refute it: Spearman −0.**232** [−0.321 ; −0.187] over 15 pairs, **+0.003** [−0.047 ; +0.068] once the two twins' individual fidelity is controlled, against the −0.7 predicted. At a distance of one, rates span 17 % to 81 %: the spread *within* a level dwarfs the spread *between* levels. Counting changed components predicts nothing useful; that *naming* them would, four one-off contrasts cannot establish. On Park the same control leaves three pairs, so no replication is possible. Two limits bound this verdict — arbitrary distance weights, and a small number of independent configurations — and are stated in the appendix.

One-factor-at-a-time from our own pipeline is arithmetically sound but not interpretable. Pipeline B and all three conditions derived from it identify the real person at chance, so each contrast opposes two noise sources: we withdraw both the reading we first gave those rates — that changing any single component collapses the channel — and the refutation of the matching preregistered prediction.

5.5 Mechanism: being right while deviating — and three refuted hypotheses

What identifies is neither the frequency of departures from the modal answer nor answer diversity, but their **correctness** — and the whole response vector is required. Correctness of deviations: 68.5 % for the human retest (leakage 81.6 %), 50.3 % for JSON 4.1 (20.6 %), 43.3 % for Demographics Only (2.14 %), 39.1 % for PMM (0.22 %) — PMM deviating and diversifying as much as the twin without leaking.

The ablation is the strongest evidence: permuting the order of each twin's 40 purchase answers drops closed-world top-1 from **33.1 % [31.2 ; 35.2] to 0.046 % [0 ; 0.11]**, below chance, and an oracle given only the count of "yes" answers gives 0.29 %. What identifies is the dependence structure between answers, not any answer in isolation. That ablation starts above this paper's headline 20.7 % because it attacks the **40 purchase items alone** where the headline uses all **60**: the 20 opinion items identify 46× less and dilute a naive Hamming rule rather than help it. That is a change of attack surface, not a discrepancy.

Three preregistered hypotheses refuted — H1 (per-item entropy drives identification), H4 (twins are more stereotyped than humans) and "deviations alone carry ≥ 80 % of the leakage": Table 1, rows 5–7, with each comparison and its interval in the appendix. We do not present "wrong deviations alone = 0.0 %" as a result: it is a floor of the method.

The effect is concentrated, and three convenient explanations are now excluded. The gap is carried entirely by the 40 purchase items: the twin re-identifies **33.2 %** [31.3 ; 35.1] of people there, against **0.08 %** [0.00 ; 0.20] for the best classical comparator explicitly fed the persona columns bearing on consumption and price; on the 20 heuristics-and-biases items the twin (0.24 %) does not exceed that comparator. This concentration is **not a content effect** — the persona names none of the 40 brands and none of the 40 products, and overlaps the purchase block's vocabulary *less* than the heuristics block's — and it is **not explained by instability of the heuristics items**, which remain reliable in humans; against the ceiling the human retest sets, the twin recovers **44 %** of the available identity information on the purchase items and **1.5 %** on the heuristics items. Token overlaps and test-retest kappas: the appendix. **The mechanism of this concentration remains unexplained**, and we propose none. *(Post-hoc, not preregistered.)*

Limits. The mechanism is established on Twin's purchase block alone and is in apparent tension with the Park archive, where opinion items identify most: at 20 items Park gives 11.7 % against 0.24 % for Twin's 20 opinion items, **at lower per-item entropy** (1.30 against 2.10). What remains is the block effect, the **best-supported hypothesis, not a result**.

5.6 Bits of identity: the instrument travels between our two datasets, the rate does not

An LLM twin costs about 0.4 bits of identity per point of accuracy gained, against 0.068 for the best useful statistical comparator. Twin JSON 4.1: **3.55 bits [3.39 ; 3.72]** of a log2 N = 11.01 ceiling. Park composite: **7.47 [7.27 ; 7.66]** of 10.04. PMM: 0.19 [0.16 ; 0.23]. B2: 0.08.

Transportability is the point. Bits normalised by human entropy give 0.0439 (Twin) against 0.0327 (Park) — a factor of **1.34** — where top-1 varies by a factor of **3.2**. Demographics Only and PMM leak with **no accuracy gain at all** (−0.5 and −2.4 points), a case a ratio alone would mask. The instrument's validity is bounded on the other side by the third dataset, where a twin carrying no demonstrable individual information has no ratio worth transporting (§ 5.8).

A nuance against our own hypothesis. The human retest has a ratio of 0.37 (Twin) and 0.44 (Park), as good as or better than the twins: the excess cost exists relative to statistical predictors, **not relative to a human answering twice**.

Limits. The bits are a **lower bound** (dyadic bound plus Miller-Madow). The sum of per-item mutual information is unusable as a total: 58.7 bits against a ceiling of 10.04 on the Park archive, as preregistered.

5.7 The risk depends on the recipe — and we could not reproduce it

Our own twins, produced in a single call to cheap models from a raw profile, identify almost nobody: 0.00 % to **0.83 % [0 ; 2.15]** (deepseek-v4/R1) across 7 cells, against 2.13 % for the other team's Demographics Only. That interval **contains** 2.13 %, so no precise factor is established — only the order of magnitude, that no model reaches 1 %. Any previously circulated "25× below" figure is withdrawn. Memorization is not the explanation: llama31-8b, with a training cutoff preceding publication, copies nothing verbatim and leaks no more than the models that do copy — a control on our own twins, in a regime where none identifies almost anyone, which we do not extend to the published twins of § 5.2–§ 5.4 (§ 1.3).

Two further preregistered predictions refuted (Table 1, rows 8–9): R1 ≥ 10 % on at least 2 of 3 models, where no model reaches 1 %; and call granularity as the explanation, which is not confirmed and is underpowered, with top-10 running **opposite** to the prediction.

We claim neither that leakage comes from the call format nor that the cause is identified. **It is not.**

We then paid to test the obvious explanation, and it failed. `openai/gpt-4.1` (Twin's own model class) on Twin's per-item recipe without modification, 30 persons × 60 items = 1,800 calls, 100 % parse rate: accuracy **0.4722 [0.4361 ; 0.5050]** against Twin's 0.574; fidelity **0.1714** against their 0.708; top-1 and top-10 **0.00 % [0 ; 11.57]** (Clopper-Pearson, n = 30). **One further preregistered prediction refuted**: accuracy > 0.55; top-1 > 5 % is **not refuted**, its interval containing the threshold — inconclusive for lack of power. We did not make the leakage come back because we did not reach Twin's fidelity, and it is that gap, not the underpowered top-1, that carries the conclusion: at the fidelity reached, the regression over the 12 points of `c7-compromis.csv` predicts **1.10 %** [−9.15 ; 11.35], and the observed 0.00 % falls inside it — from a new model *and* a new recipe, the point follows the curve.

The unknown narrows instead of disappearing. Neither the model nor the call granularity, what remains implicated is the **format and length of the profile**: Twin's full JSON persona runs to roughly 121,000 characters against our 8,000-character truncation.

Limits. n = 40 persons for the recipe arm, truncated to n = 10 on the per-item arm, and n = 30 on the paid frontier-model arm; a floor effect is not excluded. Transport failures and retries: Open Science.

5.8 A second dataset, and a third at the other end of the trajectory

On the Park archive, the composite agent designates the right participant among 1,052 **in a closed world** in **90.40 % [88.6 ; 92.2]** of cases under the strong attack, top-10 97.8 %, against a human retest ceiling of 96.8 %. **This replaces the 65.51 % [62.7 ; 68.3] we first published**: a naive agreement attack understated this archive's leakage by 38 % in relative terms. An in-sample information-weighted variant reaches 93.25 %, which we do not claim (§ 4.3).

These Park figures are not evidence that the LLM twin leaks beyond demographics. Armed with the same rarity-weighted likelihood attacker on the same basin, the demographics-only comparator reaches **85.17 %** [82.89 ; 87.26] closed-world and **29.09 %** TPR at 1 % FPR: a target-to-comparator ratio of **1.06×** [1.030 ; 1.097] closed-world — a paired gap of **+5.2 points** [+2.7 ; +8.1], statistically resolved but 27 times smaller than the 29× obtained under the naive attack — and **2.07×** at 1 % FPR. The reason is that the Park "demographics-only" condition is conditioned on **eleven** attributes that leave **98.86 %** of the 1,052 respondents unique in their own sample — a quasi-identifier block, not a demographic baseline — and that a rarity-weighted attacker converts 177 items into identity: the same comparator reaches only 3.84 % at 20 items and 22.74 % at 60. **Twin-2K-500, with 60 items and no such block, is the dataset that carries the claim** (23.23 % against 0.78 %, sixteen people, 29.9× closed-world; 4.28 % against **one person** out of 2,058 at 1 % FPR). **At that FPR we publish no ratio** — we had written 88× — by the rule this section applies to Argyle below: under 1 % the rate is fixed by the tie-breaking convention, not by the data. We defend the paired gap there. We do not withdraw Park, we requalify it: it demonstrates that linkage risk exists **without an LLM twin at all**, as soon as a quasi-identifier block and enough items are published together. A rate on this kind of corpus is therefore not comparable across studies without declaring the number **and** the identity of the items, and the tie-breaking convention. *(Post-hoc, not preregistered; audit of 13 September 2026.)*

The remaining conditions are **naive-attack measurements**, the strong attack not having been re-run on them; the highest, interview-only at **44.7 % [41.8 ; 47.5]**, carries zero documentary exposure to the survey, so it is the cleanest demonstration that an agent built from a conversation alone identifies its person — and given what the strong attack did to the composite, **it is a lower bound**. The other two conditions: the appendix.

We do not compare the Park and Twin rates as measurements of the same thing: item count, per-item entropy and human ceilings all differ. Equalised at k = 60, Park gives 34.0 % [23.5 ; 51.5] against Twin's 20.7 %, on 20 draws of a naive attack.

The third dataset, and the analysis that stops before it starts. On the Argyle archive the interpretability control **fails for all three GPT-3 twins**, and the preregistered decision rule halts the analysis there: top-1 0.14 % [0.05 ; 0.27], 0.11 % [0.04 ; 0.20] and 0.09 % [0.02 ; 0.22] against a demographic baseline of **0.10 %** [0.02 ; 0.23] recomputed on that same pool of 2,148, all intervals overlapping. No contrast is interpreted, and the anchor-pool draws were never computed: they would have interpreted noise. The verdict rests on no choice of items, dropping age or keeping only the eight attitude items leaving it unchanged (the appendix). The reason is visible per item: mean exact accuracy **47.2 %**, below the demographic imputer's 51.5 % and below the modal answer on nine items of twelve. In bits, the best of the three carries **0.094** [0.074 ; 0.118] of an 11.07 ceiling — the level of Twin's *weakest* statistical comparator — while a nearest neighbour given exactly the same eleven true answers carries **3.6 times more**; in top-1 the two are level, on a paired difference whose interval spans zero. Both sit at the noise level of this corpus.

Below 1 %, the rate is fixed by the tie-breaking convention, not by the data: across the three conventions the same twin ranges from exactly chance to more than four times the value we report, and at **matched effective information both ends of the trajectory depend on that choice**. This is why § 1.1 reports a ratio to baseline and not a bare rate. The three conventions, their rates and their intervals: the appendix.

Nor does the verdict rest on a weak attacker. Re-running the A-LLR attack that carries this paper's headline figures takes the best of the three to **0.23 %** and its baseline to **0.09 %** — still overlapping, still failing the control, the whole gain **three** people out of 2,148 (five against two). In the open-world metric that twin detects **one** person at 1 % false accusations and none at ten times that severity, at an AUC indistinguishable from zero: the metric has no shape here, which is why the left end is read closed-world. These witnesses are post-hoc.

Two preregistered predictions were missed, and we count them as neither. We had announced top-1 2.5 % [1.0 ; 6.0] and normalised bits [0.020 ; 0.070] from our measured item and pool curves; the observations, 0.14 % and 0.0044, fall far outside both. Neither our scaling law nor the transportability claim is refuted by this, because neither is testable here: both presuppose a twin that carries individual information. **We cannot distinguish "our law overstates leakage at a low effective item count" from "this**

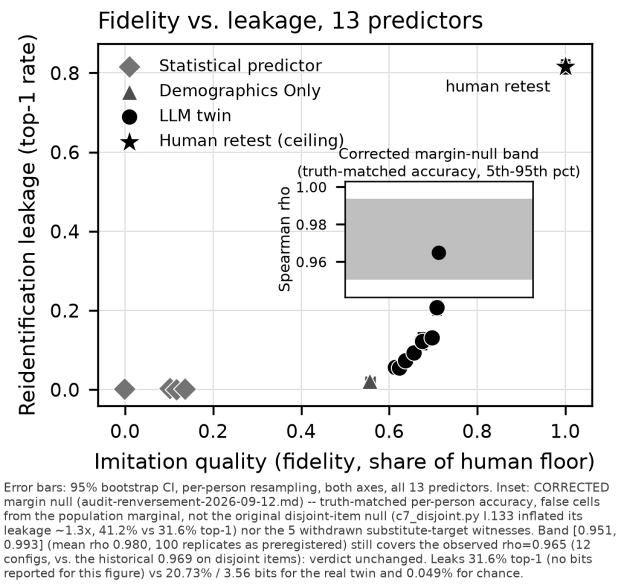

Figure 2: Imitation quality against leakage, with its control. Abscissa: individual fidelity (`fidelite_plancher`, share of the human floor). Ordinate: closed-world top-1 leakage. Thirteen points — LLM twins, statistical reference points, Demographics Only, and the human retest at fidelity ≈ 1 / leakage 81.6 % — all with error bars on both axes from a per-person bootstrap of 2,000 resamples. Inset: the 5th–95th percentile envelope of the corrected marginal null (mean rho 0.980, 100 replicates) as a grey band, with the observed rho (0.965) inside it. *At a glance*: the points rise together with no plateau at the top, but the observed correlation sits inside the band that a null matched only on accuracy already produces — the coupling is measured, not shown to be individual-specific. Data: `resultats/c7-compromis.csv`, `resultats/c7-compromis-robustesse-points.csv`, `resultats/c7-nul-corrige-marginal.csv` (inset band: `variante=marginal`, `n_replicats=100`), `resultats/c7-nul-corrige.csv` (inset point, row `rho_resume/reel`).

twin is at the noise level", and we do not pretend to. Both rows sit outside Table 1's count, like the distance law of § 5.4 — three verdicts reached after the census of § 7.1 was closed.

5.9 Scale

From N = 50 to N = 2,058, the twin/demographic ratio grows from **2.8 to 9.7** while top-1 falls from 53.2 % [50.0 ; 56.3] to 20.7 % — naive-attack rates, the scale study not having been re-run under the strong attack, and on intervals that span the sub-pool draws rather than the persons of § 4.3. **We extrapolate no value beyond N ≈ 4,000**: a power law and a logarithmic law fitted on the same 6 points already diverge at twice the range. The six pools, their rates and the two fits: the appendix.

6 Defense

Shuffling the purchase answers between people of the same demographic segment (D4) brings closed-world top-1 from **20.7 % [19.0 ; 22.4]** to **0.13 %** against the naive, non-adaptive attack, and to **0.29 % [0.10 ; 0.53]** against the best attacker we measured (§ 6.1) — the figure we report as the defense's rate. Per-item distribution and between-segment differences are preserved **exactly, by construction**, as PRAM predicts, and that exactness is also the mechanism's central limitation: shuffling item by item within a segment leaves the within-segment multiset of each item identical (1,560 of 1,560 segment × item pairs), so D4 republishes each item's within-segment histogram without noise, and an adversary who knows a segment's other members recovers the target's 40 answers by difference. **D4 offers no formal privacy guarantee.**

The cost is not a single average. By component: **0.0 points on the per-item distribution and 0.0 on group differences — which is that republication, not a cost avoided** — and **4.4 points on inter-item correlations** — a **single permutation draw, not a ten-seed average**, the highest of ten, kept as published and marked provisional. The ten-draw mean also exceeds the 4.32 points of amplitude there was to destroy, so the component is destroyed in full either way. We therefore withdraw the summary that presented D4's cost *as* a single mean of 1.47 points: two of the three components are zero by construction, so that mean divides by three an effect falling entirely on the third. And that cost is worse than the summary suggested: against the real human answers `erreur_correlations_hum` rises by a degradation of **68.1 %** — the defended twin moves *away* from the humans on correlations, so no framing in which the cost is absorbed by an error already present is correct.

Alternatives: D1 (k=10 aggregation) gives 0.55 % for 3.8 points across all three components; D2 (noise) never descends below 1 %.

6.1 The defense against a strong, then an adaptive attacker

The figure above was measured against the naive attack, which § 5.8 shows can badly understate leakage. We re-ran it against the strong attack recalibrated on the *defended* outputs, then against an attacker who knows the mechanism and which items it leaves untouched. D4 holds: top-1 goes from 0.13 % (naive) to **0.24 %** under the recalibrated strong attack, and the **adaptive** attacker plateaus at **0.29 %** (strategy S1, leaving the 20 opinion items intact; segment-invariant strategy S3 alone 0.05 %) — two orders of magnitude below the 20.7 % undefended rate, and never above 1 %. **Attribute disclosure is not demonstrated either**: the best strategy names the correct `S_gra` segment in **7.7 %** of cases, against 6.8 % at chance and **12.6 %** for always answering the most frequent segment — the attack does worse than not attacking.

A reservation that bounds the guarantee. This residue comes *entirely* from the 20 opinion items D4 does not permute: the guarantee holds for this split, not for a design leaving a more informative block untouched. On those 20 items alone the defended and the **undefended** twin are indistinguishable — 0.245 % [0.073 ; 0.471] against 0.260 %, on intervals that almost coincide — so the residue is not a residue of the defense but the part of the publication it never touches. None of these rates is a bound on the defense; they are the rates of the attacks we built.

6.2 Differential privacy: a comparison we withdraw

We preregistered that at a moderate budget differential privacy would be dominated by D4, and reported that prediction as refuted. **We withdraw the comparison and its verdict: we did not compare the cost of differential privacy to D4's.** Our reference implementation was too weak on two counts: it used the Laplace mechanism under basic sequential composition, where the Gaussian mechanism composed under zCDP (delta = 1e-6) injects **2.80 times less noise** at the same eps = 3; and it fitted the generator on the humans while scoring it against the twin, charging it the gap between those two references, worth **3.12 points** of distribution error on its own.

Corrected on both counts, over these 2,058 people and ten replicates, a synthetic generator with independent per-item marginals costs the same on distribution error and on between-segment differences whether noised at eps = 3, at eps = 10, or **not noised at all**. What we measured is therefore the cost of the independent-per-item architecture; the marginal contribution of the privacy budget is **at most 0.16 point** at eps = 3 and indistinguishable from zero at eps = 10. On correlations the DP generator preserves **none** of the amplitude to be preserved. Every figure, budget by budget, is in the appendix. D4 for its part carries **no formal guarantee**: it republishes each item's within-segment histogram exactly (**100.0 %** of segment × item pairs), so an adversary who knows a segment's other members recovers the target's 40 answers, and its zero distribution and group errors are that republication, not an advantage. **We set no top-1 rate of the two mechanisms against each other**: ours measures one fixed attacker, and the DP generator's is at chance. **We claim no superiority of D4 over differential privacy** — only a different trade-off, carrying no guarantee, against the particular attack we built.

What survives is architectural, and a theoretical point. The floor is not budgetary: the non-private control pays the same price, so a per-item independent synthesiser cannot carry this questionnaire's joint structure at any epsilon. And DP protects an individual's *membership* behind a published statistic, whereas our attack assumes the person is already known — her profile being the twin's input — and asks whether **the output conditioned on her** can be linked back: record linkage, not membership. The DP generator escapes it only by **never conditioning on an individual** (individual fidelity stays at most 0.2 points at every budget, including infinite), that is, by refusing the twin's task. A panel holder who needs population statistics should use DP; one who needs a per-person twin cannot get one here at any epsilon.

6.3 What the defense costs a downstream analyst

A defense is usable only if the analyses people actually run survive it. We measured three on the same 2,058 persons. **Identical — and that identity is the artifact, not a utility preserved**: group comparisons match to **16 decimal places** between raw and defended twin, which is the same exact republication of the within-segment histogram that makes D4's distribution and group costs zero (§ 6), not a property the mechanism preserves at a price. Demographic regression coefficients keep their sign and significance. **Destroyed**: purchase-item coefficients significant in the raw twin become non-significant after D4, and on a PCA of the 40 purchase items **45 % of the first component's loadings are inverted** relative to humans, against **0 %** for the raw twin.

The honest comparison. The unprotected twin was already wrong — it inverts the sign of the male-female gap and loses PC1+PC2 variance (the appendix) — and D4 adds **3.5 points**, less than the error already present; but the axis inversion is **D4's own doing**, changing *which* items compose the structure more than its strength. With a D4-defended twin, group comparisons are unchanged *because they are republished unchanged*, predominantly demographic regressions remain reliable, and anything resting on the link between two answers of the same person becomes unusable.

Limits. Tested on a single block, of a single twin, of a single dataset.

7 Discussion and Limitations

7.1 The thirteen refuted preregistered predictions, one withdrawn, two inconclusive, and one untestable

Seventeen predictions were registered before computation; **thirteen are refuted.** Table 1, in the appendix, gives each as written, with its verdict and source file. Eleven of the thirteen are also stated where they were measured — the headline open-world rates (§ 5.3), the quality-leakage coupling that motivated this work (§ 5.1), the twin-to-twin channel on the Park archive and on the nineteen-common-item pairs (§ 5.4), the three mechanism hypotheses (§ 5.5), our own twins and the paid frontier model (§ 5.7), and the attacker model (§ 5.3, § 6.1) — so this count can be read without turning to the table. The other two belong here: **per-item entropy does not rank items with a consistent sign across the two datasets**, and **the cost per unit of individual fidelity, which we predicted roughly constant, is not** (§ 7.2).

Two of the thirteen went in our favour, and we mark them rather than bank them: a stronger attacker gains less than the 20 % relative we predicted on Twin-2K-500, and an adaptive attacker knowing the mechanism fails to break the defense. A prediction refuted in the direction one hoped for is worth less than one refuted against.

Four verdicts left the count after they were written, each costing us a refutation. That call granularity explains the leakage, and that the frontier twin identifies more than 5 % of its cohort, were first recorded as refuted on zero-event samples; a percentile bootstrap there returns nothing but "[0 ; 0]", and the exact intervals contain the thresholds they were said to exclude: **inconclusive and underpowered** (§ 5.7). That two organisations' independent pipelines leak into one another is arithmetically answered (§ 5.4), but twins on both arms identify the real person at chance, so the contrast decides nothing: **not testable, counted as neither.** That our defense dominates differential privacy at a moderate budget is **withdrawn, not decided** — the comparison behind it was retracted (§ 6.2).

Seventeen predictions: thirteen refutations, one withdrawn, two inconclusive, one untestable. Three further outcomes were partial and count as neither (§ 6.3, § 5.7); one held outright, P2 (§ 6.1); three verdicts reached after this census closed are reported in place, not counted in it (§ 5.4, § 5.8).

7.2 What we do not know

An excess cost in bits per unit of fidelity, real but unexplained. On equal samples, CV(bits/fidelity) = 0.436 against CV(bits/accuracy point) = 0.357 over the same 9 points. With so few points and no identified mechanism, the excess **must not be attributed to "LLMs" as a family**.

We did not reproduce the leakage with our own twins (§ 5.7), including with a paid frontier model on Twin's own per-item recipe — an open limit, never a mechanism, and a narrower one than before: neither the model nor the call granularity, leaving the persona's format and length.

Whether the twin-to-twin channel crosses independent pipelines. Everything we report on that channel is measured inside a shared pipeline; the arm built to decide the cross-pipeline case fails the control of § 4.5 (§ 5.4, row 17): **T2 is untested, not refuted**.

Why the effect concentrates on one block. On Twin-2K-500 the whole gap lives in the 40 purchase items. Semantic contamination of the persona is refuted, and differential predictability between the two blocks is real but far too small to account for it (§ 5.5): **the mechanism is unexplained, and we offer no candidate.**

Rates do not predict across datasets. The normal-maxima model lands close on Park k = 60 (0.354 predicted, 0.340 observed) but fails both its controls (0.053 for 0.206 observed; 0.983 for 0.656): an isolated success between two failures is a coincidence, not a law. Only conditional information retains sign and order of magnitude across datasets (r = 0.72 and 0.89). The twin-to-twin channel's cross-dataset rate is likewise only partly explained, a residue of roughly 3× surviving both controls (§ 5.4). And nothing we hold separates the trajectory of § 5.8 from the protocols that differ along it (§ 1.1).

7.3 Multiplicity

Most of this paper's central results are bootstrap confidence intervals around a descriptive measure — re-identification rate, rank correlation, bits of identity — not classical hypothesis tests; of the 34 **adjudicated tests** underpinning those claims at the time of the census (not the 17 preregistered predictions — the appendix sets the three counts side by side), 11 were refuted and 3 judged inconclusive, roughly one third, the opposite of the signature of data dredging. **That rate controls the auxiliary family, not the headline**: the preregistered predictions concern mechanism, defense, recipe and distance law, while none of 20.7 %, 90.40 % and 60.17 % is a preregistered threshold — each is descriptive, chosen among competing metrics (top-1 or top-10, naive or strong, closed or open world, the choice of k).

Holm and Benjamini-Hochberg were both applied to the 15 tests convertible to a p-value. **Neither changes any verdict**, and the two agree on all 15: the p-values are bimodal, so no claim here rests on a marginally significant result. The intervals carrying the other claims must be read as measurements, each with its own margin, not as independent rejections of a common null. The adjusted p-values, the full 47-test census across three families, and why it and Table 1 do not carry the same totals: the appendix.

Three thresholds are fragile, and we name them rather than let a reviewer find them: the frontier twin's **fidelity > 0.10** bar, confirmed at 0.1714 but refuted had it been set at 0.18 — while our own weakest known twin sits at 0.177; the choice of k in top-k (§ 5.2); and the functional form of the scale law beyond the measured range (§ 5.9). A common bootstrap across analyses reusing the same 2,058 people was abandoned for cost, so the intervals of § 5.1–§ 5.3 on Twin carry no simultaneous coverage.

7.4 Scope, and two limits established by failure

Three datasets, one language, one questionnaire format, categorical closed-choice answers only. The mechanism analysis and the defense both rest on a single block of a single dataset. We make no claim about conversational agents or panels outside the three studied here. Two further limits we state with the resource that would lift them, because we looked for it and did not find it.

Generality beyond American attitude instruments, and the audit bottleneck. All three datasets are American survey panels built on neighbouring instrument families, so we cannot say whether this channel is a property of that family — nor, as § 1.1 states, separate the trajectory from the protocols that differ along it. The obstacle is not that the method is rare, but that the matched file — row-by-row, not by demographic cell — is almost never redistributed. We verified three such datasets at the source; we ran no systematic census, so no upper bound is established. **Not a count, a typology**: two teams withhold their file (editorial choice or silence); a third is barred by its source licence (SOEP); and one inverse case exists, synthetic outputs with no matched humans. **On that typology of four cases, and on no census, the risk appears to be created faster than it becomes auditable** — a reading of four cases, not a rate. The resource that would lift our limit is a matched individual dataset from a domain distinct from GSS and Big Five for which twins produced by a **third-party team** already exist for the same respondents, under a licence permitting local computation and publication of an aggregate rate; and, for the trajectory of § 5.8, several model generations over the same items and the same population.

Free-text outputs. We measure nothing about free text, because no dataset here permits it: the Park archive publishes **no transcripts**, **none of the 13 Twin configurations** produces text, and the Argyle twins are single recoded tokens. Every rate in this paper therefore concerns closed-choice categorical answers, and we extrapolate them to free-text outputs in neither direction — the free-text inference literature [26, 38] attacks a richer signal than ours, so our figures are not an upper bound there. The missing resource is Park's complete transcripts, or a run of the Twin pipeline that generates text.

A Provenance and validity controls

Provenance controls (§ 4.5). On Twin-2K-500, the context contains 0 wave-4 columns and 0 wave-4 QIDs; maximum normalised mutual information between context and item is 0.16 against 0.31 for the retest on the same item. On cells where wave 4 differs from waves 1–3, the JSON 4.1 output equals wave 4 in 37.7 % of cases and waves 1–3 in 37.9 % — symmetric, so no indication of copying. On the Park archive, over 34,915 cells where the human changed their mind, the composite follows wave 1 in 41.6 % and wave 2 in 43.2 %, and the 4.3–4.9 point wave-1 bias of the other three conditions holds for the `demographic` condition too, so it is a shared calibration artifact, not item leakage. On the Argyle archive each twin column is generated with its own item held out, so no tautology exists on either side; the same holds for B-demo and B-oracle, whose target item is removed from their context.

B Secondary measurements behind the results

Rank correlations, the configuration-level resampling (§ 5.1). Resampling the **configurations** — the unit the claim is about — gives **[0.749 ; 1.000]** for the first correlation, with 13.9 % of draws at exactly 1.000, and a Fisher-z approximation gives [0.848 ; 0.989]; a permutation test rejects independence at $p < 10^{-4}$. These witnesses are post-hoc.

One quantity, several runs (§ 5.2, § 5.5). Three rates in the body are the same quantity measured in different runs, differing by the tie-breaking draw alone, and each is quoted beside the term it is compared with so that both come from one run: closed-world top-1 of JSON 4.1 over the 60 items, 20.7 % canonically and 20.66 % in the comparator audit of § 5.2 (and 20.6 % in the deviations run whose bracketed rates § 5.5 quotes); and top-1 over the 40 purchase items, 33.1 % [31.2 ; 35.2] in the ablation of § 5.5 against 33.2 % [31.3 ; 35.1] in that same audit. None is substituted for another; the registry carries one identifier per run.

Three counts, and what each counts (§ 7.1, § 7.3). 17 preregistered predictions, the unit of Table 1 and § 7.1, of which thirteen are refuted; **34** adjudicated tests bearing on this article's claims, of which 11 are refuted and 3 inconclusive; **47** adjudicated tests across three families, of which those 34 are the first family. A preregistered prediction is written before computation to be able to fail; an adjudicated test is any verdict the census could settle. The refutation rates of the first two are therefore not comparable.

Closed-world comparators (§ 5.2). The six comparators of § 5.2, in order: PMM k=10, 0.23 % [0.06 ; 0.42]; B2 argmax, 0.07 % [0 ; 0.19]; k=1 donor on context alone, 0.13 %; LR on context, 0.22 %. The two individualised *generators*, sampled LR and Gaussian copula, give 0.25 % [0.11 ; 0.41] and 0.22 % [0.11 ; 0.36].

Threshold census at FPR = 0.1 % (§ 5.3). The threshold rests on a single absolute false positive on Park and two on Twin, and on Park **538 distinct thresholds** sit within 1.5 false positives of the target with TPRs spanning **0.10 % to 50.95 %**. The conservative step-function value — the largest TPR whose FPR stays at or below the target — is **50.19 %** on Park and **1.26 %** on Twin; we report no point estimate at this FPR. At FPR = 1 % the same construction is stable: 10 and 20 absolute false positives, spans of 0.38 and 0.05 point.

Twin-to-twin channel, secondary tables (§ 5.4). Demographics Only against the rich twins gives 7.76 %. The anti-artifact decoy control gives 0.04 % on the 60-item pairs and 0.10 % on the 19-item pairs. On the Park archive, top-10 is 45–46 % and the demographic bound 1.5–2.5 %. Decomposing agreement over the 60 common items: the floor between *strangers* is **46.0 %**, the demographic-ideological segment adds **3.8 points**, the person adds **17.4**. On the cross-dataset anomaly, the 177 GSS items give a real ratio of effective items of **0.89, not 2.95**, and shrinking Twin's pool to 1,052 moves top-1 from 37.3 % to 44.1 %; the same denominator places the Argyle archive at **4.31** effective items for 12 raw ones. In the T2 arm the two pipelines' own twins reach 0.0 % and 0.8 % top-1 against 0.83 % expected on a pool of 120. Two limits bound the distance-law verdict: the distance weights are arbitrary, and refitting them on the linkage rate would make the measure circular; and the independent unit is the configuration — 18 pairs, 9 configurations, two teams.

Mechanism, the three refuted hypotheses and the concentration controls (§ 5.5). H1 (entropy): opinion items carry 2.10 bits against 0.99 for purchase items and identify 46× less — top-1 0.24 % [0.07 ; 0.46] against 11.0 % [9.9 ; 12.3] for the purchase subset matched to them on entropy, a 20-item comparison. H4 (stereotypy): 36.2 % [35.6 ; 36.8] against 37.5 % [36.9 ; 38.1] in humans. "Deviations alone carry ≥ 80 % of the leakage": 0.68 % in the mixed condition. On the concentration, the persona's vocabulary overlaps the purchase block *less* (5.8 % of its distinctive tokens) than the heuristics block (38 %), and the heuristics items remain reliable in humans (median test-retest kappa 0.365, no item below 0.25, against 0.672 on the purchase items).

Recipe arm, the top-10 reversal (§ 5.7). On the call-granularity arm, top-10 runs **opposite** to the prediction: 7.5 % for the single call against 0 % for the per-item call, where top-1 is zero on both arms (Table 1, row 9).

What separates the Park and Twin rates (§ 5.8). The two archives differ in item count (177 against 60), in per-item entropy (1.30 against 0.99) and in human ceiling (96.8 against 81.6), which is why we do not read their rates as measurements of the same thing.

Scale, the six pools (§ 5.9). The share of the human ceiling falls from 57.0 % to 25.4 % between N = 50 and N = 2,058. The intervals are **not** the person-level bootstrap of § 4.3: they span the sub-pool draws, and at N = 2,058 no sub-pool is left to draw, so that row carries none. A power law and a logarithmic law fitted on the same 6 points diverge at twice the range: 18.1 % against 13.9 %.

Tie-breaking conventions and item-choice robustness on the Argyle archive (§ 5.8). Ties resolved in the attacker's favour give 0.61 % [0.33 ; 0.98], the uniform-draw expectation we report gives 0.135 % [0.045 ; 0.269], and ties resolved against the attacker give 0.047 % [0.000 ; 0.140] — exactly chance for a pool of 2,148; the leading tie class holds 13 people, median 3 candidates. Dropping age gives 0.08–0.13 % and the eight attitude items alone 0.09–0.11 %. The nearest neighbour given the same eleven true answers carries 0.342 [0.301 ; 0.389] bits and a top-1 of 0.140 % [0.075 ; 0.218], against 0.135 % [0.045 ; 0.269] for the twin — a paired difference of −0.005 points [−0.131 ; +0.137]. On the Park archive the two conditions not cited in § 5.8 are survey-only, 20.6 % [18.2 ; 23.0], and demographic, 2.26 %, both naive-attack measurements.

C Costs of the defense

Differential-privacy costs, budget by budget (§ 6.2). The generator costs **0.89 to 1.04** points of distribution error and **2.64 to 2.79** points on between-segment differences; with no privacy at all it costs 0.88 and 2.64 points. The marginal contribution of the budget at eps = 10 is 0.15. On correlations the DP generator scores **4.55 to 4.60** where the amplitude to be preserved is **4.32**.

Analyst cost, the raw twin's own error (§ 6.3). The unprotected twin inverts the sign of the male-female gap (+0.010 against −0.047 in humans) and loses 5.3 points of PC1+PC2 variance.

D Multiplicity: adjusted p-values and the full census

Multiplicity, the adjusted p-values and the full census (§ 7.3). Of the 3 tests carrying a classical p-value in the first pass, a Holm correction leaves the two smallest standing (0.002 and 0.008, adjusted to 0.006 and 0.016) and confirms the non-significance of the third. The second pass extended the confirmatory family to 15 tests convertible to a p-value and applied both Holm and Benjamini-Hochberg: the p-values are bimodal, eight at or below 0.0025 and seven already near 1. The full census covers 47 adjudicated tests across three families — the article's claims, the standalone controls, the abandoned branches — with 25 confirmed and 15 refuted; its first family, the article's claims, holds 34 of those tests, 20 confirmed, 11 refuted, 3 inconclusive. **None of these three totals is the count of preregistered predictions**, which is seventeen (Table 1): an adjudicated test is any verdict the census could settle, a preregistered prediction is one written before computation to be able to fail. The census was written over the 15 `c7-*` sub-studies existing at the time and excludes the verdicts added since, which is the second reason Table 1 and the census do not carry the same totals.

E The seventeen preregistered predictions, one by one

Ethical Considerations

Datasets, and the terms we are bound by. Twin-2K-500 is public under CC BY 4.0 and the Argyle archive under CC0 1.0. The Park replication package is publicly posted on OSF (node `t6g7k`), but public posting is not a licence: the OSF API returns `node_license: null` for that node, so **no licence is declared** and public access carries no right to redistribute the individual responses it contains. Its questionnaire content is a General Social Survey instrument, and **NORC's terms of use forbid reproducing the GSS in any form without prior written consent**. We record these constraints because an article about privacy that passes over the terms of use of its own data disqualifies itself.

What we do in consequence, verified rather than assumed. We redistribute no individual response file from any of the three archives. We checked our own released material file by file: the review artifact (Open Science) ships only code and a synthetic dataset generated at run time and reads nothing from `data/`, which is excluded from version control; every published table under `resultats/` is an aggregate by condition, by item or by bootstrap replicate; and the Argyle analysis prints and writes no ANES case identifier and no individual match. Only analysis code and disclosure-reviewed aggregates are published, with the NORC citation and attribution to Park et al. Two limits we state rather than resolve: NORC's published terms address reproduction of the GSS and do not speak to derived aggregate statistics, the reading that per-item aggregates fall outside the prohibition being ours and not NORC's; and the ANES clause covering the public variables the Argyle archive carries was not verified at its own source, the Dataverse deposit itself being CC0.

No new data was collected: the attack uses only already-published content (human answers and twin outputs). A limit to note: participants did not specifically consent to a re-identification test on the twins generated from their answers.

No individual is identified. No identifier linking a record to a real person is published or listed; only rates aggregated over the full cohorts are reported.

No new harm on Twin-2K-500. It already publishes, for each person, their identifiers and real answers beside their twin: matching is trivial without any attack. The result documents a generic risk for anyone releasing "anonymised" twins without that key — a practice Twin-2K-500 does not follow.

The Park archive is a different case, and we make no such claim there. That argument rests on a key published beside the twins and does not transfer: no trivial alignment makes the matching free there, the rates are far higher, and the objects we attack are the very ones whose individual release the authors treated as a privacy matter. What limits the harm is narrower: we publish no individual record and no per-person result from that archive, only aggregate rates, and the disclosure below is addressed to its authors, whose own warning this work quantifies.

Benefit. Several teams publish individualised LLM twin outputs without any linkage risk analysis. This provides a first measured figure for that risk on public datasets, and a reproducible method — attack plus interpretability control — for testing it *before* publication.

Responsible disclosure, and its actual state. Letters to the Twin-2K-500 authors and to Park et al. are drafted, with a 30-day response window and an offer to share code and report in advance. The tone toward Park et al. is fixed: *your warning was accurate, here is its measured magnitude.* **As of this submission the letters have not been sent and the 30-day window has not opened.** Sending them is on the critical path and is to happen before any preprint posting or code release. We write this plainly rather than imply a consultation that has not taken place: the committee should know the window was still closed when this manuscript was filed.

The Park et al. case. Their supplementary material already warned participants that their information might be "inadvertently shared" and acknowledged that complete anonymity remains challenging. They named the risk, obtained an ethics agreement worked over more than six months, pseudonymised, restricted access to part of the material and planned a 25-year withdrawal. What is nevertheless public is the replication package we analyse, which carries the 1,052 participants' individual responses together with the agent conditions' responses (§ 4.1): the protection announced in the paper does not cover the objects this attack consumes. We flag that gap as a question for the authors rather than a finding against them — from outside we cannot tell whether the restriction was lifted or never covered the replication package — and it is precisely what the disclosure letter must state correctly. No public audit had measured the magnitude of that named risk; our figures (90.40 % closed-world, 60.17 % open-world at 1 % false accusations, and 44.7 % from the interview alone, a lower bound) are, to our knowledge, its first

Table 1: The seventeen preregistered predictions and their outcomes (§7.1).

#	Prediction (preregistered)	Outcome	Source
1	The quality-leakage coupling exceeds a null matched on accuracy alone	**Refuted.** Null rho 0.980 [0.951 ; 0.993] vs 0.965 observed, 12 configurations, 100 replicates as preregistered; the original null's defect is corrected, verdict unchanged (§ 5.1)	`c7-nul-corrige-marginal.csv`
2	Open-world > 5 % (Twin) and > 30 % (Park) at FPR = 1 %	**Refuted as first measured** (3.04 % and 20.39 %). Under the strong attack Twin still fails at 4.28 %, Park **passes** at 60.17 % — but that attack also takes Park's demographic comparator to 29.09 % (§ 5.8). Half refuted, half confirmed, one row	`c7-attaquant-fort-resultats.md`
3	Twin-to-twin top-1 ≥ 20 % on the Park archive	**Refuted.** 11.9–13.0 % at 177 common items (failure criterion not met either)	`c7-transfert-stanford-resultats.md`
4	The twin-to-twin channel leaks on the 19-common-item pairs	**Refuted.** 0.45 % mean top-1, at chance (0.05–0.10 %)	`c7-transfert-resultats.md`
5	H1: per-item entropy drives identification	**Refuted.** Opinion items 2.10 bits, identify 46× less	`c7-mecanisme-resultats.md`
6	H4: twins are more stereotyped than humans	**Refuted.** 36.2 % [35.6 ; 36.8] vs 37.5 % [36.9 ; 38.1]	`c7-mecanisme-resultats.md`
7	Deviations alone carry ≥ 80 % of the leakage	**Refuted.** 0.68 % in the mixed condition	`c7-deviations-resultats.md`
8	R1 ≥ 10 % on ≥ 2 of 3 of our own models	**Refuted.** No model reaches 1 %	`c7-gen-resultats.md`
9	Call granularity explains the leakage	**Inconclusive, underpowered.** 0.00 % both arms ([0 ; 8.8] at n=40, [0 ; 30.85] at n=10); top-10 runs opposite	`c7-recette-resultats.md`
10	Cost per unit of individual fidelity is roughly constant	**Refuted at equal sample.** CV 0.436 vs 0.357 on the same 9 points	`c7-compromis-resultats.md` § 5
11	Per-item entropy correlates with identifying power consistently	**Refuted.** Opposite sign by dataset: r = −0.81 (Twin), +0.57 (Park)	`c7-bits-resultats.md` § 3
12	P1: a stronger attacker gains ≥ 20 % relative over the naive attack	**Refuted on Twin** (+12.2 %, 20.7 → 23.23 %); held on Park (+38.0 %, 65.51 → 90.40 %)	`c7-attaquant-fort-resultats.md`
13	P3: an adaptive attacker knowing the mechanism breaks the defense	**Refuted, for the defense.** Plateaus at 0.29 %, never above 1 %	`c7-attaquant-fort-resultats.md`
14	At a moderate budget, DP is dominated by our defense on the aggregate table	**Withdrawn, not decided.** The comparison is retracted: the DP generator was fitted on the humans and scored against the twin, and at eps = ∞ — no privacy — the cost was already the same (§ 6.2)	`audit-comparaison-dp-2026-09-13.md`
15	A frontier model on Twin's per-item recipe reaches accuracy > 0.55	**Refuted.** 0.4722 [0.4361 ; 0.5050]	`c7-fort-resultats.md`
16	That same twin reaches top-1 > 5 %	**Inconclusive.** 0.00 % [0 ; 11.57] (Clopper-Pearson, n = 30); the 5 % threshold lies inside it	`c7-fort-resultats.md`
17	T2, independent pipelines: top-1 CI excludes the segment control and stays ≥ 2× the demographic baseline	**Not testable — counted as neither.** 1.76 % [0.35 ; 3.63] at n = 142/200, but both arms' twins identify the real person at chance: the contrast decides nothing (§ 5.4)	`c7-deux-organisations-resultats.md`

quantification — with the qualification established in § 5.8, that an identically armed demographics-only comparator reaches 85.17 % on this archive, so what is quantified is the linkage risk the release carries and not a margin the LLM agent adds — and our own first measurement understated it.

Publishing the attack code. *For*: replicability, and enabling other panel holders to test their twins before publication. *Against*: it lowers the cost for an attacker already holding a third-party panel's real answers and a corresponding twin file. *Our resolution*: publish, because the code only acts if the attacker already holds real answers to the same items, and the countermeasure is straightforward once the risk is known.

Recommended defenses. (a) Do not publish twin outputs item by item for blocks with high inter-person specificity; publish aggregates or calibrated noise. (b) Break the person-twin alignment in public files. (c) Publish regenerated twins without copying wave metadata (`StartDate`/`EndDate`/`Duration`/`RecordedDate`). (d) Run a minimal linkage test, and the interpretability control of § 4.5, before any twin release: both are free, and the second says whether the first can be interpreted at all. (e) Within-segment shuffling (§ 6), with its real cost stated: 4.4 points on inter-item correlations — a single permutation draw, and above the 4.32 points of amplitude there was to destroy, so the component is destroyed in full (§ 6) — a 68.1 % worsening of the gap to human correlations (5.775 → 9.709), and the loss of every inter-item analysis downstream (§ 6.3) — and with what it does not provide: **no formal guarantee**, and exact republication of each item's within-segment histogram, so an adversary who knows a segment's other members recovers the target's answers. It holds at 0.29 % against an attacker who knows the mechanism, but only for a split that shuffles the informative block (§ 6.1). (f) Differential privacy where the published object can be a population statistic rather than a per-person twin; we make no claim about its cost relative to (e), having withdrawn the comparison we preregistered (§ 6.2).

Public interest. The AAPOR report of 8 May 2026 [35] ranks synthetic response generation as the most risky of the core tasks it evaluates and names re-identification by linkage without quantifying it. A risk named by professional bodies but never measured justifies an independent measurement followed by responsible disclosure rather than silence.

Ethics review and scope. This study was not submitted to an external ethics panel such as an IRB. It relies exclusively on secondary data already made public by their original publishing teams, and no new data was collected from any person: we surveyed no one, interviewed no one, and had no contact with any data subject. On that basis we judged the study out of scope for human-subjects review, with the limit recorded above — participants did not specifically consent to a re-identification test on the twins generated from their answers. The scoping does not extend to the released artifact, which processes no real person's data at all (Open Science).

Open Science

Review artifact. `artefact/` replays the re-identification attack and its D4 defense in a single command (`./artefact/run.sh`), **with no real data and no real person**, in about 25 seconds on a laptop, without GPU, network, or any language model call. It runs on a fictional dataset of 600 synthetic persons at fixed seed with a tunable individual-signal dial, structurally comparable to the real data (categorical items, product × price matrix block, demographic segments, retest) but with different sample sizes and signal strength. It demonstrates that the two mechanics are the ones running in the paper: `attaque.py` and `defense.py` import their functions directly from `analyses/c7_reidentification.py` and `analyses/c7_defense.py`, nothing is copied, the only function that reads real data is never called, and a guard refuses to start if a path to `data/` is detected.

No artifact figure is a result of this study. The artifact's outputs (on fictional data: 21.9 % before defense, 0.842 % after D4) illustrate the mechanics and must never be cited as study results.

Underlying analyses. Every quantitative claim in this paper corresponds to a dated analysis script and result file under `resultats/` in this repository, organised to mirror the paper's own section numbering. Individual file names are cited where a reader needs them to retrace a number: in the Source column of Table 1, in the `Data:` line of each figure caption, and at the few points in § 5.1 and § 5.7 where a specific script line is at issue; elsewhere they are omitted to keep the manuscript readable.

Bibliographic verification. Of our 50 references, 23 carry an auditable source-verification trace in the artifact; the remaining 27 were verified in an earlier pass whose record did not survive. We state this as a limit rather than claim a fully audited bibliography. Full accounting in `article/references-verification.md`.

Transport failures and retries (§ 5.7). Roughly 1,490 calls first failed with HTTP 402 from in-flight credit reservation, not lack of balance, and were retried at decreasing concurrency without double billing.

Differential-privacy implementation (§ 6.2). The synthesiser was written by hand rather than with a reference library, and is not full PrivBayes: per-item marginals perturbed by Laplace noise (L1 sensitivity 2, budget split over 60 items, basic sequential composition) then sampled i.i.d. per item, that is degree-0 PrivBayes with no joint structure. It was fitted on the human answers and scored against the twin. Epsilon covers only the published histograms, with no composition across the repository's other analyses, and correlations, between-segment differences and the `S_gra` covariate are not protected. These are reasons the comparison of § 6.2 is withdrawn rather than corrected here; the corrected design — refitting on the twin, a composed Gaussian mechanism under zCDP, and at least ten seeds with intervals — is specified in `resultats/audit-comparaison-dp-2026-09-13.md`, section 9, and **has since been run** (`analyses/c7_dp_zcdp.py`, `resultats/c7-dp-zcdp.csv`, ten replicates per budget): it is that run, not the one described in this paragraph, that supplies every figure of § 6.2. The withdrawal stands regardless, and for a reason the rerun does not lift: what the corrected run measures is the cost of our own reference implementation's independence assumption, which is not a ranking of the two mechanisms in either direction.

Obtaining the real data. Twin-2K-500: Hugging Face repository `LLM-Digital-Twin/Twin-2K-500`, CC BY 4.0. Argyle et al.: Harvard Dataverse `doi:10.7910/DVN/JPV20K`, CC0 1.0. Park replication archive: OSF `https://osf.io/t6g7k/`, which carries real individual responses, is not redistributed here, and is never read by the artifact — and whose two obstacles public availability does not lift: no declared licence, and NORC's terms on the GSS content (Ethical Considerations). Obtain all three from their original sources and verify current terms of use before any download.

Preregistrations. `resultats/c7-preenregistrement.md`, `resultats/c7-defense-preenregistrement.md`, `resultats/c7-stanford-preenregistrement.md`, `resultats/c7-argyle-preenregistrement.md` and `resultats/c7-loi-distance-preenregistrement.md`, each written before any computation. Third-party timestamping is pending.

Environment. Python 3.13.14, numpy 2.5.2, pandas 3.0.5, scikit-learn 1.9.0. No network dependency.

AI Use

Large-language-model-based agents were used throughout this project: they wrote the analysis code, ran the experiments, drafted this manuscript, and conducted adversarial review of both the code and the text. We state this in full rather than narrowly, because an account that understated that involvement would mislead a reader more than a plain one. The responsible investigator directed the work throughout: set the research questions and the falsifiable predictions in advance of each analysis, decided what to keep, discard, or redo, authorized all paid computation, and is responsible for the accuracy and integrity of the submission. No generative AI tool is listed as an author.

Three checks bear on how much trust the model-produced outputs warrant. First, quantitative predictions were preregistered in writing, dated, before the corresponding analysis script was run.

Second, the analysis pipeline fixes a random seed at every stochastic step, and independent reruns are checked to reproduce identical output before a result is reported. Third, the manuscript and code were subjected to adversarial review passes whose default posture is rejection; these passes are how two substantive defects were found and corrected before submission: a statistical control meant to carry only each person's accuracy margin that in fact copied their true individual answer with the complementary probability, and a bootstrap confidence interval reported as "[0, 0]" on a zero-event sample, since replaced with an exact Clopper-Pearson interval. These checks reduce, but do not remove, the risk that a model-produced number or claim is wrong; the responsible investigator remains accountable for every figure and statement in this paper regardless of how it was produced.

References

[1] Tim Adams, Colin Birkenbihl, Karen Otte, Hwei Geok Ng, Jonas Adrian Rieling, Anatol-Fiete Näher, Ulrich Sax, Fabian Prasser, and Holger Fröhlich. 2025. On the Fidelity versus Privacy and Utility Trade-Off of Synthetic Patient Data. *iScience* (2025). doi:10.1016/j.isci.2025.112382

[2] Meenatchi Sundaram Muthu Selva Annamalai, Andrea Gadotti, and Luc Rocher. 2024. A Linear Reconstruction Approach for Attribute Inference Attacks against Synthetic Data. In *33rd USENIX Security Symposium*. arXiv:2301.10053

[3] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. 2023. Out of One, Many: Using Language Models to Simulate Human Samples. *Political Analysis* 31, 3 (2023), 337–351. doi:10.1017/pan.2023.2

[4] Vamshi Krishna Bonagiri, Juan Nicolas Sepulveda-Arias, Abdoul Jalil Djiberou Mahamadou, and Monojit Choudhury. 2026. Cognitive Digital Twins: Ethical Risks and Governance for AI Systems That Model the Mind. In *AAAI/ACM Conference on AI, Ethics, and Society (AIES)*. arXiv:2606.23094

[5] Mark Bun, Jonathan Ullman, and Salil Vadhan. 2014. Fingerprinting Codes and the Price of Approximate Differential Privacy. In *46th Annual ACM Symposium on Theory of Computing (STOC 2014)*. arXiv:1311.3158 Full version in SIAM Journal on Computing.

[6] Jessup Byun, Xiaofeng Lin, Joshua Ward, and Guang Cheng. 2025. Risk In Context: Benchmarking Privacy Leakage of Foundation Models in Synthetic Tabular Data Generation. In *Agentic & GenAI Evaluation Workshop @ KDD 2025*. arXiv:2507.17066

[7] Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. 2023. Quantifying Memorization Across Neural Language Models. In *The Eleventh International Conference on Learning Representations (ICLR 2023)*. arXiv:2202.07646

[8] Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, Alina Oprea, and Colin Raffel. 2021. Extracting Training Data from Large Language Models. In *30th USENIX Security Symposium*. arXiv:2012.07805

[9] Debeshee Das, Jie Zhang, and Florian Tramèr. 2024. Blind Baselines Beat Membership Inference Attacks for Foundation Models. In *Data Problems for Foundation Models (DATA-FM) Workshop @ ICLR 2025*. arXiv:2406.16201 Also presented at IEEE DLSP Workshop 2025.

[10] Yves-Alexandre de Montjoye, Laura Radaelli, Vivek Kumar Singh, and Alex Pentland. 2015. Unique in the Shopping Mall: On the Reidentifiability of Credit Card Metadata. *Science* 347, 6221 (2015), 536–539. doi:10.1126/science.1256297

[11] Jörg Drechsler and Anna-Carolina Haensch. 2024. 30 Years of Synthetic Data. *Statist. Sci.* (2024). arXiv:2304.02107

[12] Michael Duan, Anshuman Suri, Niloofar Mireshghallah, Sewon Min, Weijia Shi, Luke Zettlemoyer, Yulia Tsvetkov, Yejin Choi, David Evans, and Hannaneh Hajishirzi. 2024. Do Membership Inference Attacks Work on Large Language Models?. In *Conference on Language Modeling (COLM)*. arXiv:2402.07841

[13] Cynthia Dwork, Adam Smith, Thomas Steinke, Jonathan Ullman, and Salil Vadhan. 2015. Robust Traceability from Trace Amounts. In *IEEE 56th Annual Symposium on Foundations of Computer Science (FOCS 2015)*. 650–669. doi:10.1109/FOCS.2015.46

[14] Peter Eckersley. 2010. How Unique Is Your Web Browser?. In *Privacy Enhancing Technologies (PETS 2010) (LNCS, Vol. 6205)*. doi:10.1007/978-3-642-14527-8_1

[15] Vitaly Feldman. 2020. Does Learning Require Memorization? A Short Tale about a Long Tail. In *52nd Annual ACM SIGACT Symposium on Theory of Computing (STOC 2020)*. arXiv:1906.05271

[16] Georgi Ganev. 2024. Synthetic Data, Similarity-based Privacy Metrics, and Regulatory (Non-)Compliance. In *2nd Workshop on Generative AI and Law (GenLaw) @ ICML*. arXiv:2407.16929

[17] Georgi Ganev and Emiliano De Cristofaro. 2025. The Inadequacy of Similarity-based Privacy Metrics: Privacy Attacks against "Truly Anonymous" Synthetic Datasets. In *2025 IEEE Symposium on Security and Privacy*. arXiv:2312.05114

[18] Georgi Ganev and Emiliano De Cristofaro. 2026. Rethinking Anonymity Claims in Synthetic Data Generation: A Model-Centric Privacy Attack Perspective. (2026). arXiv:2601.22434

[19] Matteo Giomi, Franziska Boenisch, Christoph Wehmeyer, and Borbála Tasnádi. 2023. A Unified Framework for Quantifying Privacy Risk in Synthetic Data. *Proceedings on Privacy Enhancing Technologies* 2023, 2 (2023), 312–328. doi:10.56553/popets-2023-0055

[20] Steven Golob, Sikha Pentyala, and Martine De Cock. 2026. SoK: Reconstruction Attacks on Synthetic Tabular Data (Insights from Winning the NIST CRC). (2026). arXiv:2606.08372

[21] J. M. Gouweleeuw, P. Kooiman, L. C. R. J. Willenborg, and P.-P. de Wolf. 1998. Post Randomisation for Statistical Disclosure Control: Theory and Implementation. *Journal of Official Statistics* 14, 4 (1998), 463–478. https://www.scb.se/contentassets/ca21efb41fee47d293bbee5bf7be7fb3/post-randomisation-for-statistical-disclosure-control-theory-and-implementation.pdf

[22] Florent Guépin, Matthieu Meeus, Ana-Maria Cretu, and Yves-Alexandre de Montjoye. 2023. Synthetic is all you need: Removing the Auxiliary Data Assumption for Membership Inference Attacks against Synthetic Data. In *Data Privacy Management (DPM) Workshop @ ESORICS 2023*. arXiv:2307.01701

[23] Florimond Houssiau, James Jordon, Samuel N. Cohen, Owen Daniel, Andrew Elliott, James Geddes, Callum Mole, Camila Rangel Smith, and Lukasz Szpruch. 2022. TAPAS: a Toolbox for Adversarial Privacy Auditing of Synthetic Data. In *NeurIPS 2022 Workshop on Synthetic Data for Empowering ML Research (SyntheticData4ML)*. arXiv:2211.06550

[24] Jingchen Hu and Claire McKay Bowen. 2023. Advancing Microdata Privacy Protection: A Review of Synthetic Data. (2023). arXiv:2308.00872

[25] James Jordon, Lukasz Szpruch, Florimond Houssiau, Mirko Bottarelli, Giovanni Cherubin, Carsten Maple, Samuel N. Cohen, and Adrian Weller. 2022. *Synthetic Data – what, why and how?* Technical Report. The Royal Society. arXiv:2205.03257 https://royalsociety.org/-/media/policy/projects/privacy-enhancing-technologies/Synthetic_Data_Survey-24.pdf

[26] Myeongseob Ko, Jihyun Jeong, Sumiran Singh Thakur, Gyuhak Kim, and Ruoxi Jia. 2026. From Weak Cues to Real Identities: Evaluating Inference-Driven De-Anonymization in LLM Agents. In *International Conference on Machine Learning (ICML)*. arXiv:2603.18382

[27] Simon Lermen, Daniel Paleka, Joshua Swanson, Michael Aerni, Nicholas Carlini, and Florian Tramèr. 2026. Large-Scale Online Deanonymization with LLMs. (2026). arXiv:2602.16800

[28] Matthieu Meeus, Florent Guépin, Ana-Maria Creţu, and Yves-Alexandre de Montjoye. 2023. Achilles' Heels: Vulnerable Record Identification in Synthetic Data Publishing. In *Computer Security – ESORICS 2023*. arXiv:2306.10308 doi:10.1007/978-3-031-51476-0_19

[29] Arvind Narayanan and Vitaly Shmatikov. 2008. Robust De-anonymization of Large Sparse Datasets. In *2008 IEEE Symposium on Security and Privacy*. 111–125. doi:10.1109/SP.2008.33

[30] Joon Sung Park, Carolyn Q. Zou, Jonne Kamphorst, Niles Egan, Aaron Shaw, Benjamin Mako Hill, Carrie Cai, Meredith Ringel Morris, Percy Liang, Robb Willer, and Michael S. Bernstein. 2024. LLM Agents Grounded in Self-Reports Enable General-Purpose Simulation of Individuals. (2024). arXiv:2411.10109 v1 title: Generative Agent Simulations of 1,000 People.

[31] Tianyi Peng, Melanie Brucks, George Gui, Daniel J. Merlau, and Olivier Toubia. 2026. Digital Twins Are Funhouse Mirrors: Five Systematic Distortions. *Science Advances* 12, 36 (2026), eaeh8260. doi:10.1126/sciadv.aeh8260 22 co-authors; arXiv preprint 2509.19088 under a different title.

[32] Michael Platzer and Thomas Reutterer. 2021. Holdout-Based Fidelity and Privacy Assessment of Mixed-Type Synthetic Data. (2021). arXiv:2104.00635

[33] Luc Rocher, Julien M. Hendrickx, and Yves-Alexandre de Montjoye. 2019. Estimating the Success of Re-Identifications in Incomplete Datasets Using Generative Models. *Nature Communications* 10 (2019), 3069. doi:10.1038/s41467-019-10933-3

[34] Luc Rocher, Julien M. Hendrickx, and Yves-Alexandre de Montjoye. 2025. A Scaling Law to Model the Effectiveness of Identification Techniques. *Nature Communications* 16 (2025), 347. doi:10.1038/s41467-024-55296-6

[35] David M. Rothschild, Jenny Marlar, Ashley Amaya, Soubhik Barari, Trent Buskirk, Curtiss Cobb, Jen Gennai, D. Sunshine Hillygus, Ramya Korlakai Vinayak, Masha Krupenkin, Sunghee Lee, Darby Steiger, and Brock Webb. 2026. *Responsible AI Integration in Survey Research.* Technical Report. American Association for Public Opinion Research (AAPOR). https://aapor.org/wp-content/uploads/2026/05/Responsible-AI-Integration-In-Survey-Research.pdf

[36] Masoumeh Shafieinejad, D. B. Emerson, Behnoosh Zamanlooy, Elaheh Bassak, Fatemeh Tavakoli, Sara Kodeiri, Marcelo Lotif, and Xi He. 2026. On Privacy Leakage in Tabular Diffusion Models: Influential Factors, Attacker Knowledge, and Metrics. (2026). arXiv:2605.06835

[37] Geoffrey Smith. 2009. On the Foundations of Quantitative Information Flow. In *Foundations of Software Science and Computational Structures (FoSSaCS 2009) (LNCS, Vol. 5504)*. 288–302. doi:10.1007/978-3-642-00596-1_21

[38] Robin Staab, Mark Vero, Mislav Balunović, and Martin Vechev. 2024. Beyond Memorization: Violating Privacy Via Inference with Large Language Models. In *The Twelfth International Conference on Learning Representations (ICLR)*. arXiv:2310.07298

[39] Theresa Stadler, Bristena Oprisanu, and Carmela Troncoso. 2022. Synthetic Data – Anonymisation Groundhog Day. In *31st USENIX Security Symposium*. 1451–1468. arXiv:2011.07018

[40] Jennifer Taub, Mark Elliot, Maria Pampaka, and Duncan Smith. 2018. Differential Correct Attribution Probability for Synthetic Data: An Exploration. In *Privacy in Statistical Databases (PSD 2018) (LNCS, Vol. 11126)*. 122–137. doi:10.1007/978-3-319-99771-1_9

[41] Olivier Toubia, George Z. Gui, Tianyi Peng, Daniel J. Merlau, Ang Li, and Haozhe Chen. 2025. Twin-2K-500: A Dataset for Building Digital Twins of Over 2,000 People Based on Their Answers to Over 500 Questions. (2025). arXiv:2505.17479

[42] Joshua Ward, Xiaofeng Lin, Chi-Hua Wang, and Guang Cheng. 2025. Synth-MIA: A Testbed for Auditing Privacy Leakage in Tabular Data Synthesis. (2025). arXiv:2509.18014

[43] Zexi Yao, Nataša Krčo, Georgi Ganev, and Yves-Alexandre de Montjoye. 2025. The DCR Delusion: Measuring the Privacy Risk of Synthetic Data. In *Computer Security – ESORICS 2025*. arXiv:2505.01524 doi:10.1007/978-3-032-07884-1_24

[44] Samuel Yeom, Irene Giacomelli, Matt Fredrikson, and Somesh Jha. 2018. Privacy Risk in Machine Learning: Analyzing the Connection to Overfitting. In *IEEE 31st Computer Security Foundations Symposium (CSF 2018)*. arXiv:1709.01604

[45] Jie Zhang, Debeshee Das, Gautam Kamath, and Florian Tramèr. 2024. Membership Inference Attacks Cannot Prove that a Model Was Trained On Your Data. In *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML 2025)*. arXiv:2409.19798 Position paper.